GCP → IBM Cloud Resource Mapping

Assignment: GCP project with two GKE clusters, two web apps, multi-pod deployment & full observability — mapped to this IBM Cloud implementation.

Equivalent Direct functional equivalent **Close match** Same concept, different model **Different approach** IBM Cloud solves it differently

Not required Not applicable / not needed

1 · PROJECT STRUCTURE & GOVERNANCE

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
Resource hierarchy (Folder → Project)	GCP Project / Folder	**Resource Group** Close match	Single `niraj-sandbox-nprd` resource group scopes all resources. Looked up (not created) by `modules/governance` via `terraform-ibm-modules/resource-group/ibm`.
IAM roles for Dev, Ops, SRE, CI/CD	GCP IAM Roles & Bindings	**IAM Access Groups + Service IDs** Equivalent	Four Access Groups: `poc-k8s-rbac-dev`, `-sre`, `-ops`, `-automation`. Each has a Service ID member and scoped IKS platform + service roles. IAM Service roles map to Kubernetes RBAC via IKS (Writer → edit, Manager → cluster-admin).
Workload Identity (pod → GCP SA mapping)	GKE Workload Identity	**IBM Trusted Profiles** Equivalent	Two Trusted Profiles per cluster: `tp-apps` (SecretsReader on Secrets Manager, linked to `web-app-a` & `web-app-b` SAs in `mydemo` NS) and `tp-logs-agent` (Sender on Cloud Logs). No long-lived API keys mounted.
Infrastructure as Code	Terraform (same)	**Terraform (IBM Cloud provider)** Equivalent	`terraform/` root + 4 modules: `governance/`, `regional/`, `icd-redis/`, `transit-gateway/`. Provider pin: IBM-Cloud/ibm ≥ 2.5.

2 · NETWORKING

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
VPC with segregated subnets	VPC Network + Subnets	**IBM VPC + Subnets** Equivalent	One VPC per region via `terraform-ibm-modules/landing-zone-vpc/ibm`. Non-overlapping CIDRs: `us-south` `192.168.0.0/26`, `us-east` `192.168.0.64/26`. No public gateways — all outbound via private endpoints.

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Shared VPC / cross-region connectivity	Shared VPC / VPC Peering	IBM Transit Gateway Equivalent	Global Transit Gateway (`poc-tgw`) connects both regional VPCs. Enables Istio east-west failover (DestinationRule retries web-app-b in the remote region via Transit GW).
Cloud NAT (outbound egress)	Cloud NAT	Private service endpoints Different approach	Workers reach IBM services (ICR, ICD Redis, KMS, SM, Logs, Monitoring) directly over IBM's private backbone — no public internet egress required. No NAT gateway needed.
Firewall rules for node pools	VPC Firewall Rules	VPC Network ACLs + Security Groups Equivalent	ACL on each VPC subnet: allow inbound HTTPS (443) and HTTP (80), allow inbound TCP ephemeral return (1024–65535), allow all outbound. IBM-internal and VPC-connectivity rules prepended by landing-zone-vpc module.
Private Service Access for Google APIs	Private Service Connect / PSA	Virtual Private Endpoints (VPE) Equivalent	IKS module auto-creates VPEs for the cluster API server, container registry, and master. ICD Redis is consumed via its private-only endpoint, keeping all data traffic inside IBM's network.
Cloud DNS	Cloud DNS	IBM Cloud Internet Services DNS Equivalent	DNS for `mydemo.cistest-load.com` is managed by CIS (Cloudflare). GLB CNAME record maps the hostname to the active pool origins.

3 · GKE CLUSTER ARCHITECTURE

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
GKE Cluster (Standard/Autopilot)	GKE	IBM Kubernetes Service (IKS) Equivalent	Two single-AZ IKS clusters via `terraform-ibm-modules/base-iks-vpc/ibm` : `poc-us-south-3-iks` (AZ us-south-3) and `poc-us-east-3-iks` (AZ us-east-3). 3 workers each, flavour `bxf.4x16` , Ubuntu 24.
Node Pool — general purpose	GKE Node Pool	IKS Worker Pool Equivalent	Single default worker pool per cluster. `workers_per_zone = 3` , `machine_type = bxf.4x16` .
Separate ingress / system pool	GKE Node Pool (system)	Istio Gateway on default pool Close match	Istio managed add-on is installed on the same worker pool. The Istio Gateway acts as the edge ingress controller rather than a separate node pool.
Active/Active cross-region	Multi-cluster GKE	Dual-region IKS + Istio east-west Equivalent	CIS GLB routes to the nearest healthy pool (active/active). Within each region, Istio DestinationRule with ServiceEntry

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			retries web-app-b calls to the remote cluster via Transit Gateway on failure.
Anthos Service Mesh / Istio	Anthos Service Mesh (ASM)	IKS Managed Istio add-on Equivalent	Istio installed via `ibm_container_addons` resource. PeerAuthentication enforces mTLS STRICT across `mydemo` namespace. AuthorizationPolicy restricts web-app-b to web-app-a SA only.
etcd secret encryption at rest	CMEK on GKE (Cloud KMS)	IKS KMS config + IBM Key Protect Equivalent	`kms_config` block on `base-iks-vpc` wires the existing Key Protect root key to IKS, encrypting all Kubernetes secrets at rest in etcd.

4 · APPLICATION DEPLOYMENT

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
Deployment / ReplicaSet	Kubernetes Deployment	Kubernetes Deployment (Helm) Equivalent	Both apps deployed via Helm charts to `mydemo` namespace. Web App A: replicas=2; Web App B: replicas=2. Helm manages ConfigMaps, Secrets, ServiceAccounts.
Horizontal Pod Autoscaler	HPA (CPU / custom metrics)	HPA (CPU) Equivalent	Web App A HPA: minReplicas=2, maxReplicas=10, CPU target=60%. `POST /spike` endpoint burns CPU to trigger scale-out for demonstration.
ConfigMaps & Secrets	ConfigMap + Secret Manager	ConfigMap + IBM Secrets Manager Equivalent	App config in Helm ConfigMaps. Sensitive values (API_SECRET, Redis credentials) stored in Secrets Manager; pods read via `SecretsReader` Trusted Profile — no static secret mounts at deploy time.
StatefulSet / Cloud SQL / MemoryStore	GKE + Cloud SQL or MemoryStore Redis	IBM Cloud Databases for Redis (ICD) Equivalent	Single shared ICD Redis instance (`poc-us-south-3-redis` , standard plan, 2 members) in us-south. Both regional Web App B pods connect via TLS over private endpoint. No data-replication lag — one store serves both regions.
Artifact Registry / Container Registry	Artifact Registry / GCR	IBM Container Registry (ICR) Equivalent	Shared ICR namespace `ndp-demo` . `deploy.sh` builds and pushes both images; both clusters pull via `private.us.icr.io` private endpoint. Authenticated via `ibmcloud cr login` .

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Binary Authorization (image attestation)	Binary Authorization	Not implemented (ICR Vulnerability Advisor available) Not required	ICR includes Vulnerability Advisor for image scanning. Binary Authorization-equivalent policy enforcement is not configured in this implementation.

5 · CUSTOMER TRAFFIC & GLOBAL LOAD BALANCING

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
Global External HTTPS Load Balancer	Google Cloud Load Balancer (GCLB)	CIS Global Load Balancer (GLB) Equivalent	CIS GLB at `mydemo.cistest-load.com` performs active/active routing across two pools: `us-south-web-app-a` and `us-east-web-app-a`.
Multi-cluster Ingress / NEGs	Multi-Cluster Ingress (MCI) + NEGs	CIS GLB health-check pools Close match	Each CIS pool targets one IKS cluster's Istio Gateway LoadBalancer IP. Health check: `GET /status`. Pool unhealthy → GLB automatically fails all traffic to the remaining pool.
SSL termination at edge	GCLB SSL policy + Cloud Armor	CIS TLS termination + WAF/DDoS Equivalent	CIS (Cloudflare) terminates TLS at the edge, applies WAF rules and DDoS protection before traffic reaches any IBM datacenter.
GKE Ingress Controller	GKE Ingress / NGINX / ASM Ingress	Istio IngressGateway Equivalent	Istio Gateway resource + VirtualService routes inbound traffic from the CIS origin to the correct Service in the `mydemo` namespace.
Kubernetes Service (ClusterIP / NodePort)	Kubernetes Service	Kubernetes Service (ClusterIP) Equivalent	Helm charts create ClusterIP Services for both apps. Istio sidecar proxies handle pod-level load-balancing, retries, and circuit-breaking.
Internal load balancer for east-west	Internal HTTPS Load Balancer / MCS	Istio DestinationRule + ServiceEntry + Transit GW Close match	`k8s/istio/web-app-b-east-west.yaml` — DestinationRule with two subsets (local weight 100, remote weight 0). On local failure, Istio retries to the remote cluster's VIP via the Transit Gateway private network.

6 · OBSERVABILITY (LOGGING, MONITORING, TRACING)

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
Container & cluster logging	Cloud Logging (GKE logging agent)	IBM Cloud Logs Equivalent	Single shared instance (`poc-us-south-3-logs` , standard plan). Logs agent (Helm, `terraform-ibm-modules/logs-agent/ibm`) deployed to both clusters; ships to the shared instance via private ingress endpoint.

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			Auth via Trusted Profile (`tp-logs-agent` , Sender role) — no static API key.
Metrics (CPU, HPA, LB, 5xx)	Cloud Monitoring	IBM Cloud Monitoring Equivalent	Single shared instance (`poc-us-south-3-monitoring` , lite plan). Monitoring agent (Helm, `terraform-ibm-modules/monitoring-agent/ibm`) deployed to both clusters; uses shared `access_key` with private endpoint only. Covers pod CPU/memory, HPA events, Istio telemetry.
Distributed tracing	Cloud Trace	Istio telemetry (partial) Close match	Istio sidecar collects distributed trace headers (Zipkin/Jaeger compatible). Full Cloud Trace-equivalent (IBM Instana or OpenTelemetry export) is not explicitly configured but compatible.
CPU/memory profiling	Cloud Profiler	Not implemented Not required	IBM Instana provides continuous profiling for containerised workloads if required. Not in scope for this implementation.
Error Reporting	Cloud Error Reporting	IBM Cloud Logs (log-based alerts) Close match	Application exception logs flow into IBM Cloud Logs. Log-based alerts and views can filter on severity ERROR/CRITICAL to replicate error aggregation.
BigQuery + Grafana dashboards	BigQuery + Managed Grafana	IBM Cloud Monitoring dashboards Close match	IBM Cloud Monitoring ships with a built-in Grafana-compatible UI. Pre-built IKS dashboards cover pod restarts, request latency percentiles, CPU/memory utilisation, and HPA scale events — equivalent to the 4 required panels.
Prometheus / managed Prometheus	Managed Prometheus (GMP)	IBM Cloud Monitoring (Sysdig) Equivalent	IBM Cloud Monitoring is built on Sysdig, which is Prometheus-compatible. The monitoring agent scrapes standard `/metrics` endpoints and forwards to the shared Monitoring instance.

7 · SECURITY

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
Secret Manager	Secret Manager	IBM Secrets Manager Equivalent	Pre-existing Secrets Manager instance. Registered as cluster Ingress SM. App pods read secrets via `SecretsReader` Trusted Profile — no static secrets injected into pods at deploy time.
Cloud KMS (CMEK)	Cloud KMS	IBM Key Protect Equivalent	Pre-existing Key Protect instance. Root key wired to both clusters via `kms_config` on `base-iks-vpc` for

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			Kubernetes secret (etcd) encryption at rest. ICD Redis uses provider-managed encryption.
Cloud Armor (WAF)	Cloud Armor	CIS WAF + DDoS protection Equivalent	IBM Cloud Internet Services (Cloudflare) provides WAF rules and DDoS scrubbing at the edge before traffic enters IBM Cloud.
Mutual TLS between services	Anthos Service Mesh mTLS	Istio PeerAuthentication STRICT Equivalent	`k8s/istio/peer-authentication.yaml` enforces mTLS STRICT across the `mydemo` namespace. All pod-to-pod traffic is encrypted and mutually authenticated.
AuthorizationPolicy (RBAC)	ASM AuthorizationPolicy	Istio AuthorizationPolicy Equivalent	`helm/web-app-b/templates/authz-policy.yaml` — web-app-b accepts traffic only from the `web-app-a` ServiceAccount. Any other source is denied at the Envoy proxy layer.
Private GKE clusters	Private GKE (no public API endpoint)	IKS private-only endpoint Equivalent	`disable_public_endpoint = true` on both clusters. Terraform and `deploy.sh` require private network access (VPN or IBM Cloud Shell) to reach the API server.
Non-root container security	Pod Security Standards / PSP	Kubernetes securityContext (Helm) Equivalent	Both Helm charts set `runAsNonRoot: true`, drop all Linux capabilities, and set a read-only root filesystem where applicable.

8 · HIGH AVAILABILITY & DISASTER RECOVERY

GCP CONCEPT	GCP SERVICE	IBM CLOUD EQUIVALENT	IMPLEMENTATION DETAIL
Cross-regional GKE redundancy	GKE multi-region clusters	Dual IKS clusters (us-south + us-east) Equivalent	Active/active: CIS GLB routes to both pools simultaneously. Either cluster can independently serve all traffic.
Multi-cluster ingress failover	Multi-Cluster Ingress	CIS GLB pool health checks Equivalent	Each pool's `GET /status` health check detects cluster unavailability within seconds. GLB stops routing to the unhealthy pool automatically.
Cross-regional DB (Cloud SQL HA / Memorystore replication)	Cloud SQL HA / Memorystore cross-region replication	Single shared ICD Redis (us-south) Close match	A single ICD Redis instance in us-south serves both regions. Cross-region consistency is automatic (no replication lag) at the cost of higher latency from us-east. For stricter HA, a second ICD Redis in us-east with application-level sync can be added.
GKE etcd backups	GKE managed etcd backups	IKS managed control plane backups Equivalent	IBM fully manages IKS control plane (etcd). Customer is responsible for

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			application-level backups via Velero or similar.

SUMMARY

> **Coverage:** All 8 sections of the GCP assignment are addressed in the IBM Cloud implementation. Every core GCP service has a direct or close functional equivalent. The two key design differences are: (1) IBM Trusted Profiles replace GCP Workload Identity with a keyless model that requires no secret injection at pod startup; and (2) CIS (Cloudflare-backed) provides the global load balancer and WAF in a single service, replacing the combination of GCLB + Cloud Armor + Cloud DNS. The observability gap (Cloud Trace / Cloud Profiler / BigQuery-backed Grafana) can be closed by adding IBM Instana or an OpenTelemetry collector to the existing IBM Cloud Monitoring setup.

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